Quanser AERO – FINAL PROJECT
AE 443 Experimental D&C
Embry-Riddle Aeronautical University

Part 1: Qualitative PID Control

1 Background

The Quanser Aero has two thruster modules, each of which is driven by a DC motor. The motor armature circuit schematic for one of the thrusters is shown in Figure 1.1 and the electrical and mechanical parameters are given in Table 1.1. The DC motor shaft is connected to the propeller hub. The hub is a collet clamp used to mount the propeller to the motor and has a moment of inertia of J_h. A propeller is attached to the output shaft with a moment of inertia of J_p.

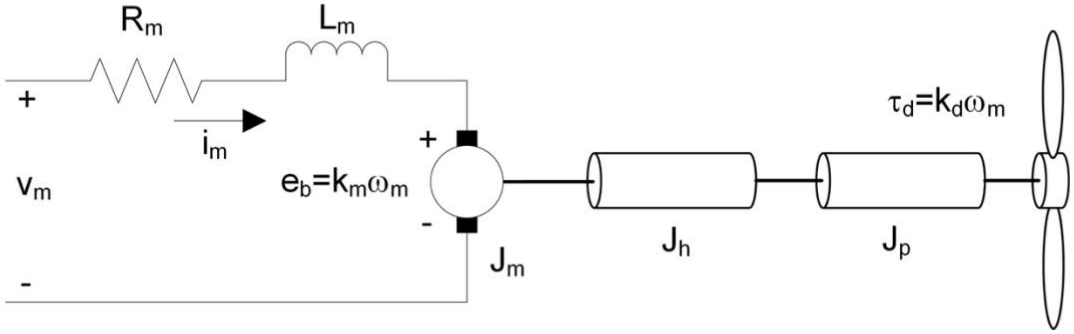


Figure 1.1: Quanser Aero thruster motor and load

The back-emf (electromotive) voltage $e_b(t)$ depends on the speed of the motor shaft, ω_m, and the back-emf constant of the motor, k_m. It opposes the current flow. The back emf voltage is given by:

$$e_b(t) = k_m \omega_m(t) \tag{1.1}$$

The torque exerted by drag and air resistance has been simplified into an experimentally derived coefficient of the speed of the propeller. In this model the drag torque τ_d which opposes the motor torque is given by:

$$\tau_d(t) = k_d \omega_m(t) \tag{1.2}$$

Symbol	Description	Value
DC Motor		
R_m	Terminal resistance	$8.4\ \Omega$
k_t	Torque constant	$0.042\ \mathrm{N \cdot m/A}$
k_m	Motor back-emf constant	$0.042\ \mathrm{V/(rad/s)}$
J_m	Rotor inertia	$4.0 \times 10^{-6}\ \mathrm{kg \cdot m^2}$
L_m	Rotor inductance	$1.16\ \mathrm{mH}$
Propeller		
k_d	Drag/Air resistance coefficient	$1 \times 10^{-5}\ \mathrm{N \cdot m/(rad/s)}$
J_h	Propeller hub inertia	$3.04 \times 10^{-9}\ \mathrm{kg \cdot m^2}$
J_p	Propeller inertia	$7.2 \times 10^{-6}\ \mathrm{kg \cdot m^2}$

Table 1.1: Quanser Aero system parameters

Using Kirchoff's Voltage Law, we can write the following equation:

$$v_m(t) - R_m i_m(t) - L_m - \frac{di_m(t)}{dt}\ k_m \omega_m(t) = 0. \tag{1.3}$$

Since the motor inductance L_m is much less than its resistance. Then, the equation becomes

$$v_m(t) - R_m i_m(t) - k_m \omega_m(t) = 0. \tag{1.4}$$

Solving for $i_m(t)$, the motor current can be found as:

$$i_m(t) = \frac{v_m(t) - k_m \omega_m(t)}{R_m}. \tag{1.5}$$

The motor shaft equation is expressed as

$$J_{eq} \omega_m(t) = \tau_m(t) \tag{1.6}$$

where J_{eq} is total moment of inertia acting on the motor shaft and τ_m is the applied torque from the DC motor. Based on the current applied, the torque is

$$\tau_m = k_m i_m(t) \tag{1.7}$$

1.1 Torques Acting on the Quanser Aero

The free-body diagram of a Quanser Aero in the 1-DOF configuration that pivots about the pitch axis is shown in Figure 1.1.

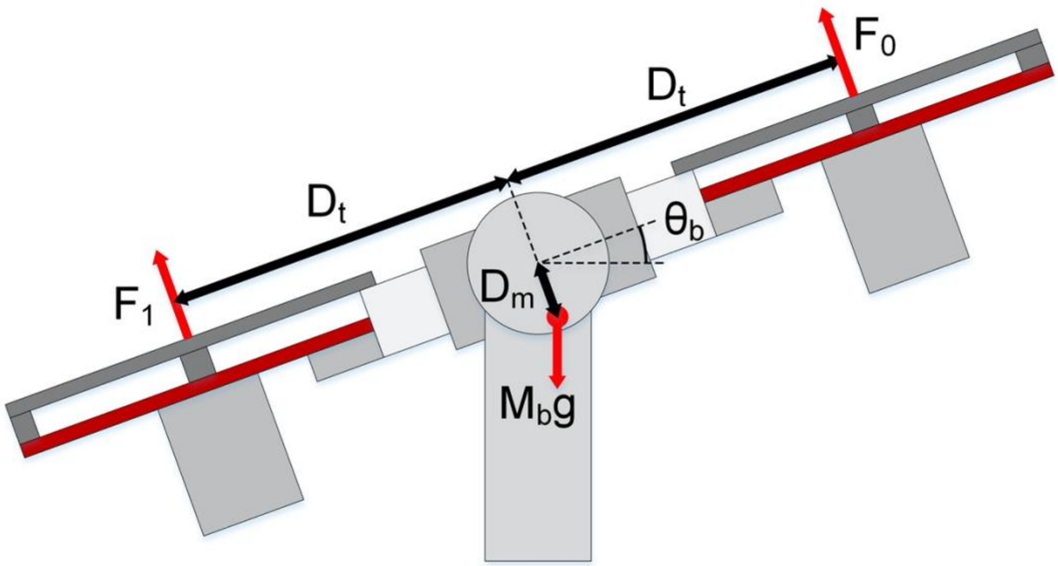


Figure 1.2: Free-body diagram of 1-DOF Quanser Aero

Symbol	Description	Value
M_b	Aero body mass	1.15 kg
D_t	Thrust displacement	0.158 m
D_m	Center of mass displacement	0.0071 m

Table 1.1: Quanser Aero system parameters

The torques acting on the rigid body system at rest can be described by the equation

$$\tau_t - M_b g(D_m \sin\theta_b) = 0 \tag{1.8}$$

where D_m is the distance below the plane of the Aero body of the center of mass as depicted in Figure 1.1. The thrust torque is given by

$$\tau_t = F_0 D_t - F_1 D_t \tag{1.9}$$

where F_0 and F_1 are the thrust forces perpendicular to the aero body generated by thrusters 0 and 1 respectively. Thus the complete torque equation becomes

$$F_0 D_t - F_1 D_t - M_b g(D_m \sin\theta_b) = 0. \tag{1.10}$$

The forces exerted by the thrusters are related to the applied voltage by the formula

$$F = K_t V \tag{1.11}$$

where Kt is the thrust coefficient of the thruster. Since the motors act in opposition to each other in the same degree of freedom, the motor voltages V_0 and V_1 can be replaced with a single control variable:

$$V_m = V_0 - V_1 \tag{1.12}$$

Thus with respect to the difference in current applied to the motors, the torque equation becomes

$$K_t V_m D_t - M_b g(D_m \sin\theta_b) = 0 \tag{1.13}$$

1. Proportional Compensation

A proportional compensator drives the plant based on the difference between the current position of the system and the desired position. This difference is magnified by the *proportional gain* k_p which can be found either experimentally or calculated based on system requirements such as rise time. Greater proportional gain will result in a system with a shorter *rise time*, that is, the time needed for the system to reach the desired position. However since the system is constantly accelerating toward the set point, large proportional gains will generally lead to a system with large overshoot, and which oscillate for a long time.

1.2 Derivative Compensation

To deal with the overshoot and oscillation caused by a proportional compensator, many systems implement a derivative compensator in parallel. This compensator drives the plant based on the rate of change of the position (or velocity) of the system. As with proportional control, this derivative is magnified by the *derivative gain* k_d. The derivative compensator effectively acts as added damping in underdamped systems. To improve the stability of systems with derivative compensation, filtering is often added to prevent spikes in the derivative component of the compensation due to signal noise.

1.3 Integral Compensation

In many cases, the combination of proportional and derivative gain will result in a system which does not settle sufficiently close to the setpoint. In this case, an integral compensator may be added. This compensator drives the system based on the integral of the error over time magnified by the *integral gain* k_i. This component of the controller increases the longer the system remains far from the setpoint.

1.4 PID Control

The complete proportional, integral, and derivative control can be expressed mathematically as follows

$$u(t) = k_p e(t) + k_i \int_0^t e(\tau)d\tau + k_d \frac{de(t)}{dt}. \tag{1.14}$$

The corresponding block diagram is given in Figure 1.1. The control action is a sum of three terms referred to as proportional (P), integral (I) and derivative (D) control gain. The controller Equation 1.1 can also be described by the transfer function

$$C(s) = k_p + \frac{k_i}{s} + k_d s. \tag{1.15}$$

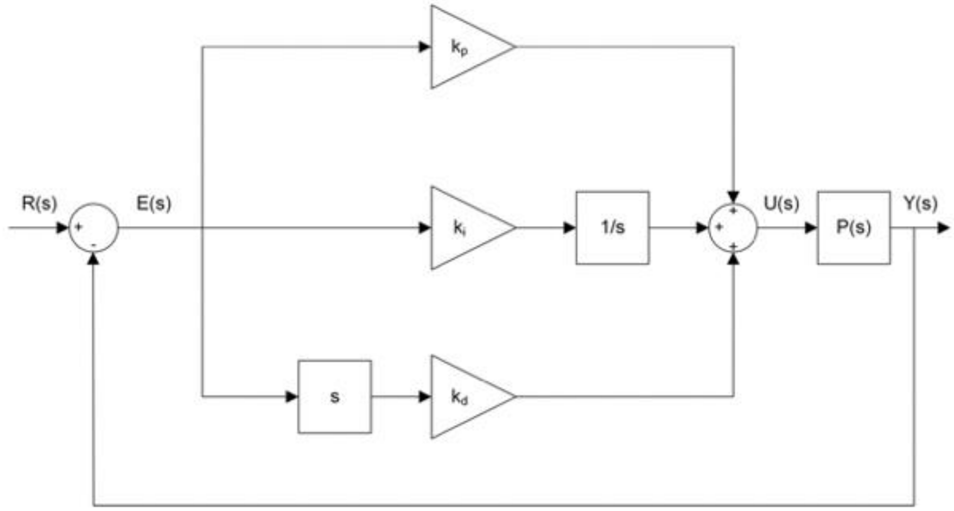


Figure 1.3: Block diagram of PID control

2 In-Lab Exercises

2.1 Open Loop Response

1. Open **q_aero_PID_specification.mdl**. As shown in Figure 2.1 This QUARC® model implements basic command and unity feedback for the Quanser Aero. A control signal of 0 fed into the Motor Voltage Gain will command the Quanser Aero to remain at "Level" flight, that is, with +17V applied to both motors. The command signal is given a gain of 7 and added and subtracted from the "Level" voltage for motor 0 and 1 respectively such that over the full command range of ± 1 the difference in command voltages between the motors ranges from +14V to -14V while both thrusters remain in their linear operating range of 10-24V.

   Delete the signal between the mux port and the saturation and add a new mux element. Connect a step input and a constant to both inputs of the mux element. Configure the step with an initial value of 10 and a final value of 15. Save the response and measure peak time, settling time for 5%, frequency, damping, and PO.

2.2 Proportional Control

1. Open **q_aero_PID_specification.mdl**. As shown in Figure 2.1 This QUARC® model implements basic command and unity feedback for the Quanser Aero.

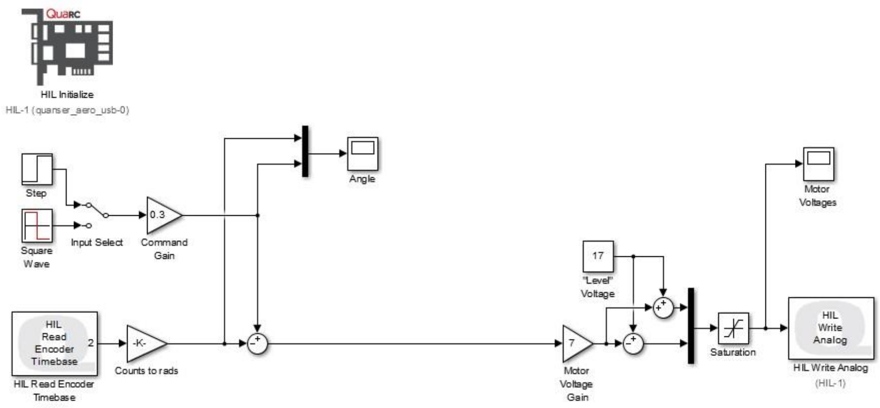
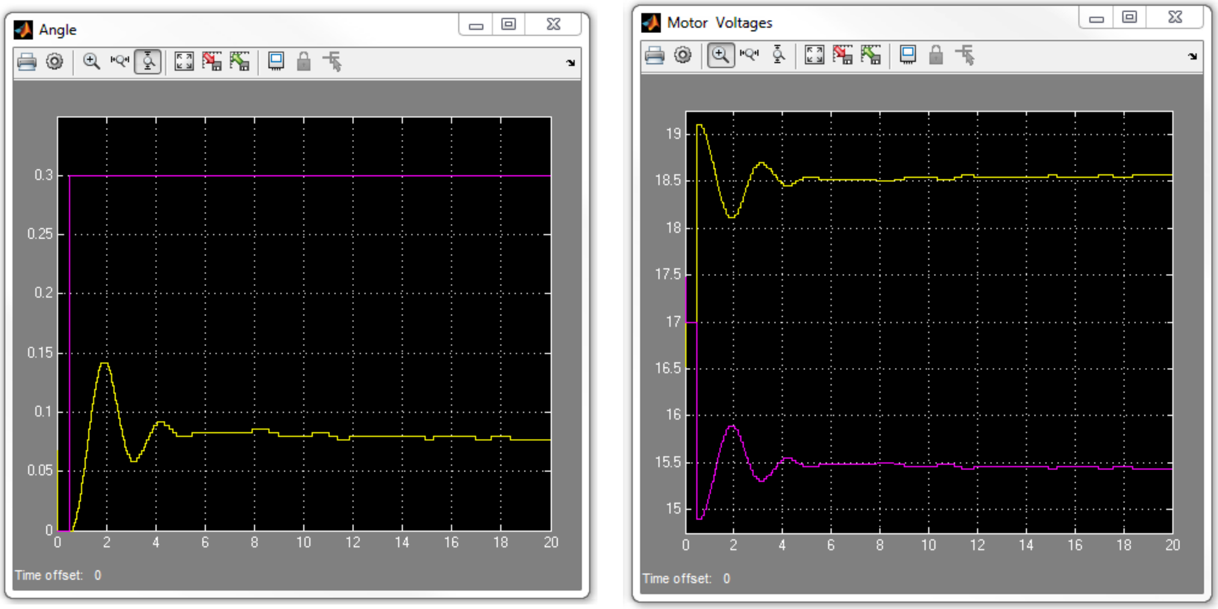


Figure 2.1: Unity-feedback block diagram

2. Build and run the QUARC® controller. The response should look **similarly** as shown in Figure 2.2.



(a) Angle

(b) Voltage

Figure 2.2: Quanser Aero unity feedback step response.

3. Add a proportional gain block kp between the error sum and the Motor Voltage Gain as shown in Figure 2.3

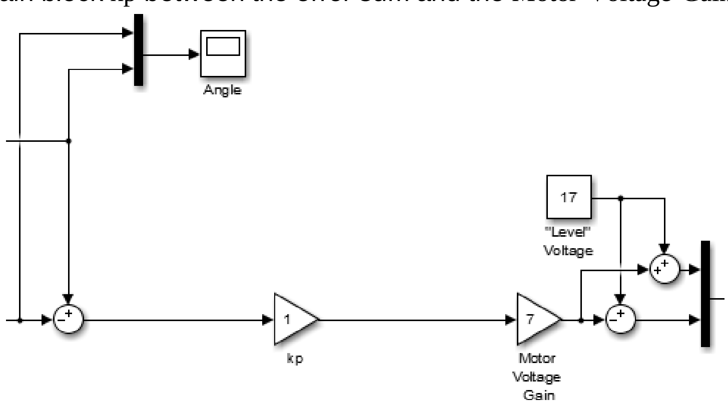


Figure 2.3: Proportional gain in feedback loop

4. Vary the proportional gain between 5 and 10 Run the QUARC® model with various proportional gains. What effects does increasing the proportional gain have on the system response?

5. Given the specification that the controlled system has a peak time of less than 1.1 seconds, is it practical to use purely proportional control? Why or why not?

2.3 Derivative Control

1. A transfer function and derivative gain block k_d has been added in parallel with the proportional gain as shown in Figure 2.4.

2. Because the angle data from the encoder is a discrete signal, some filtering will have to be added to the derivative to smooth out discontinuities when the encoder count changes. Set the transfer function for the derivative to

$$F(s) = \frac{20s}{s+20} \tag{2.1}$$

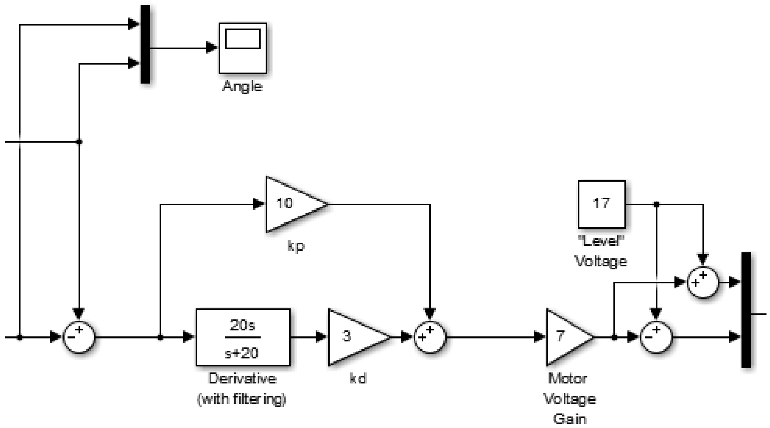


Figure 2.4: PD gains in feedback loop

3. Set the proportional gain to 10 and vary the derivative gain between 4 and 8. Run the QUARC® model with various derivative gains. What effects does increasing the derivative gain have on the system response? How does the system response differ when the derivative gain is very small (e.g. <0.25)
4. Change the Input Select switch to select the square wave input.
5. It is desired that the PD compensated system has a peak pitch which does not exceed ±0.35 radians when commanded with a 0.3 radian square wave. What is the minimum value of k_d which produces a system with acceptable overshoot?

2.4 Integral Control

1. The response should now approach the commanded value quickly and settle in a short time. However even with a large proportional gain, the final position of the Aero still differs from the command value by approximately 0.5 radians to combat this steady-state error, add an integrator with an integral gain k_i in parallel with the other two gains as shown in Figure 2.5.

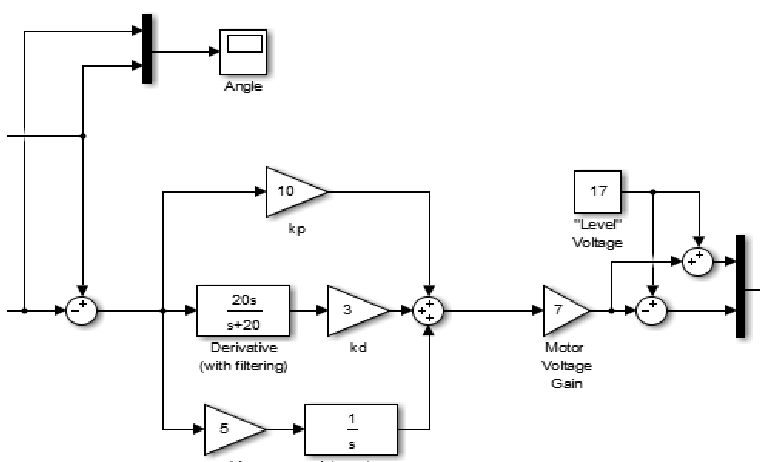


Figure 2.5: PID gains in feedback loop

2. Keeping the proportional and derivative gain at the values identified in step 5 vary the integral gain between 3 and 6. Run the QUARC® model with various integral gains. What effects does increasing the integral gain have on the system response? What happens when the integral gain is increased too much?

3. Assume the specification that the system settles within 3% of the command value. For the 0.3 radian square wave this means that the steady state value must be in the range from 0.291 to 0.309 radians. What integral gain results in a system which meets this requirement?

2.5 Response Tuning

1. If your response does not match the overshoot (<0.35 radians peak pitch) and peak time (<1.1 seconds) specifications from previous steps, try tuning your control gains until your response satisfies all three requirements (including steady-state error). Plot the resulting system response overlaid on the command waveform and linear representations of the peak overshoot limits. Record the final PID gains and comment on how you modified your controller to arrive at those results.

2. Stop the QUARC® controller.

Part 2: PID Control to Specification

3 Background

3.1 Second-Order System Model

For the purpose of this lab we will consider the Aero under the closed loop unity feedback control to be a second-order system with a transfer function of the form

$$\frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \tag{3.1}$$

where ω_n is the natural undamped frequency and ζ is the damping ratio. The properties of its response depend on the values of the ω_n and ζ parameters. Consider when a second-order system is subjected to a unit step. The system response is shown in Equation 3.1, where the blue trace is the response (output) $y(t)$ and the red trace is the step input $r(t)$.

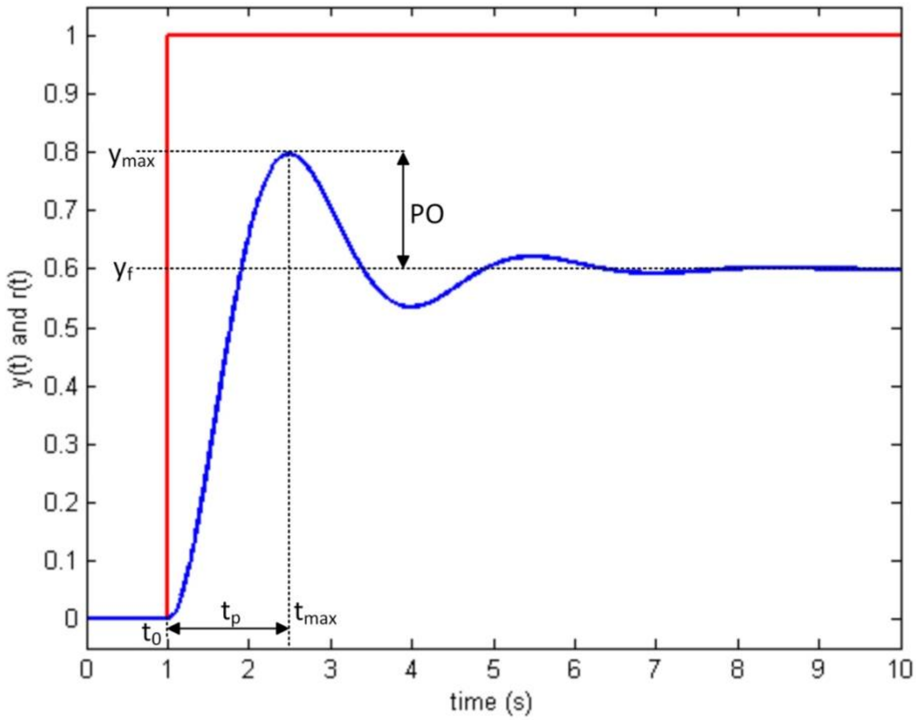


Figure 3.1: Response of a second order system to a unit step

The maximum value of the response is denoted by the variable y_{max} and it occurs at a time t_{max}. For a response similar to Figure 3.1, where the system settles at a final value of y_f the percent overshoot is found using

$$PO = 100\left(\frac{y_{max} - y_f}{y_f}\right). \tag{3.2}$$

In a second-order system, the amount of overshoot depends solely on the damping ratio parameter and it can be calculated using the equation

$$PO = 100e^{\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right)}. \tag{3.3}$$

From the initial step time t_0, the time it takes for the response to reach its maximum value is

$$t_p = t_{max} - t_0. \tag{3.4}$$

This is called the peak time of the system and it depends on both the damping ratio and natural frequency of the system. It can be derived analytically as

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}. \tag{3.5}$$

Generally speaking, the damping ratio affects the shape of the response while the natural frequency affects the speed of the response.

One other useful measure of system response is how long the system takes to settle. This is defined as the time required for the system to settle within a certain percentage δ of y_f. Since the second order response is bounded by $e^{-\zeta\omega_n t}$ this occurs at approximately

$$e^{-\zeta\omega_n T_s} = \delta. \tag{3.6}$$

3.2 Controlling a second-order system

Consider a general second order system with the open loop transfer function

$$P(s) = \frac{a}{s^2 + bs + c}, \tag{3.7}$$

and a general PID controller described by the transfer function

$$C(s) = k_p + \frac{k_i}{s} + k_d s. \tag{3.8}$$

where k_p is the proportional gain, k_i is the integral gain, and k_d is the derivative gain. The compensated system has the combined open-loop transfer function

$$L(s) = \frac{a(k_d s^2 + k_p s + k_i)}{s(s^2 + bs + c)}, \tag{3.9}$$

and the unity feedback closed-loop characteristic equation

$$L(s) = s^3 + (b + ak_d)s^2 + (c + ak_p)s + ak_i. \tag{3.10}$$

This equation can be set equal to a prototypical third-order closed loop characteristic equation

$$L(s) = (s + p_0)(s^2 + 2\zeta\omega_n s + \omega_n^2), \tag{3.11}$$

where p_0 is the position of a zero and ω_n and ζ describe the desired response. The following equalities can be determined by gathering equal powers of s

$$k_d = \frac{2\zeta\omega_n + p_0 - b}{a}, \tag{3.12}$$

$$k_p = \frac{\omega_n^2 + 2\zeta\omega_n p_0 - c}{a}, \tag{3.13}$$

$$k_i = \frac{\omega_n^2 p_0}{a}, \tag{3.14}$$

which give required gains for the PID controller.

4 In-Lab Exercises

4.1 Second-order system approximation

Modeling the Quanser Aero precisely would require a third- or even fourth-order system model. However qualitative design of a compensator for such high-order systems is generally prohibitively complex. Instead, the Quanser Aero will be subjected to a step input and the response will be used to generate a second-order transfer function which approximates system behavior.

1. Open **`q_aero_PID_specification.mdl`**.
2. Configure the PID controller so that there is a unity feedback loop. This can be accomplished by setting the K_p gain block to one, and the K_d and K_i gain blocks to zero.
3. Build the QUARC® controller. Make sure the Input Select switch is set to the Step function, set the Command gain to 1 and run the model. The response should look **similar** to that shown in Figure 4.1.

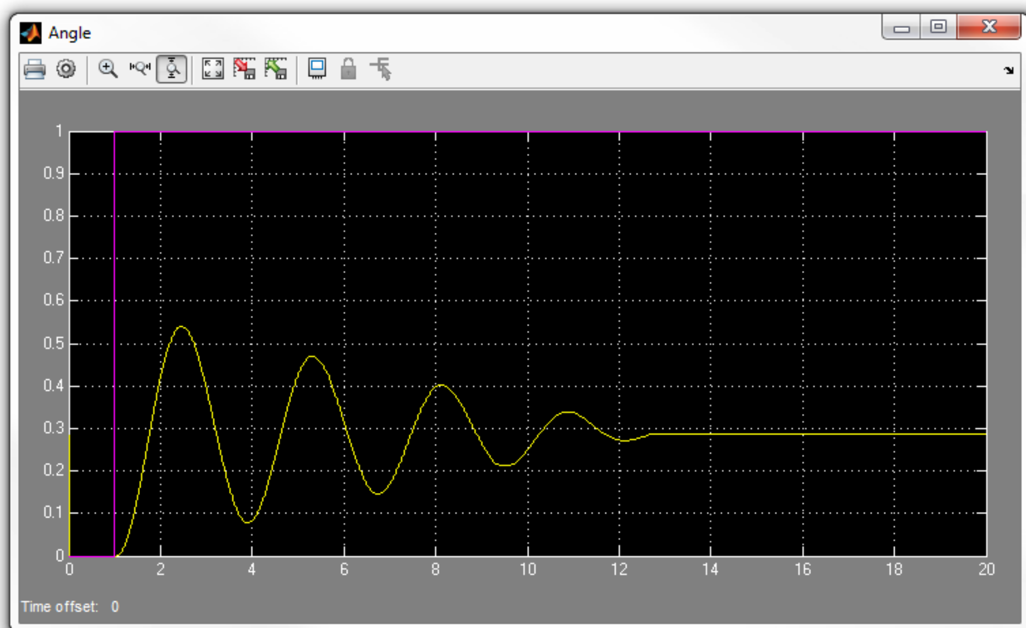


Figure 4.1: Response of the aero to a unit step with unity feedback

4. Analyze the plotted angle data to find t_p, y_{max}, y_f and the peak overshoot.
5. Using Equation 3.3 to Equation 3.5 it is possible to find the values of ω_n and ζ which define a second-order system with a similar rise behavior (Solution to be used in the experiment: ω_n = 2.213 rad/s, ζ = 0.0227)
6. Because $y_f \neq 1$ in response to a unit step, a slightly modified closed-loop transfer function is required. Using the prototypical transfer function and the values calculated in step 5 find the transfer function of the second-order approximation of the Quanser Aero plant.

$$\frac{Y(s)}{R(s)} = \frac{y_f \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2(1 - y_f)}, \tag{4.1}$$

7. Change the values in the Model Transfer Function block to match the calculated second-order approximation.

8. Build and run the QUARC® model. How closely does the second-order approximation model the behavior of the Quanser Aero plant? What physical dynamics in the Quanser Aero might contribute to the difference between the model and the plant behavior?

4.2 Calculating PID Gains

1. The desired response for the compensated Aero system is a critically damped system with a 5% settling time of 2 seconds. Given that critical damping requires $\zeta = 1$, equation 3.5 provides a natural frequency of approximately $\omega_n = 2$ rad/s. Using these values along with the values for a, b, and c from the second-order transfer function in step 6 and a zero position $p_0 = 1$, calculate the PID gains using the formulas given in the background section. Solution to be used in the experiment: K_p = 3.3739, K_i = 3.06, K_d = 3.7484

2. Set the Model Enable switch to zero since the model will not be needed for testing the controller.

3. The normal operating range of the Quanser Aero is ±0.5 radians. Adjust the command gain value to 0.3 and change the input select switch to the square wave.

4. Update the control gains to match those found in step 1.

5. Build and run the QUARC® model. Observe the response of the compensated system to the square wave. For each of the steps, does the compensated system meet the requirements for damping and settling time?

6. Stop the QUARC® controller.

7. Power *OFF* the Quanser Aero.

5 Home Exercises

5.1 Frequency Analysis

1. Using the same values from 4.2.1 to model the system with equation 4.1, Plot a bode diagram for the open-loop system, and a second bode diagram for the close-loop system that includes the designed PID controller. Calculate phase margin, gain margin, bandwidth, and compare both open-loop and close-loop and describe how the PID changed the system.

5.2 LQR Control

1. Using the same values from 4.2.1 to model the system with equation 4.1, Find the state-space model of the system, then verify if it's controllable and observable and explain what this means. If the system is controllable and observable, design an LQR controller (Linear Quadratic Regulator controller where the input $u = -kx$). Is this control a good approach? (Hint: Compare with the voltage limit applied to the motors of the Quanser AERO and the Voltage of the PID experiment. You can use the conversion implemented in the Quanser Simulink) Include a screenshot of your Simulink implementation.

6 Final Report

I. PROCEDURE
1. **Briefly** describe main goal and procedures of Section 2
2. **Briefly** describe main goal and procedures of Section 4

II. RESULTS
1. Open Loop Response for section 2.1.
2. Tuned PID values for section 2.5.
3. Tuned PID values for section 4.2.
4. Response plot for tuned PID in section 2.5.
5. Comparison between model and Transfer Function in section 4.1
6. Response plot for tuned PID in section 4.2.
7. Table with values for section 2.1

III. ANALYSIS
1. Compare and discuss the differences between both methods. Do the gains differ much between methods (section 2.5 and 4.2)? Show calculations for section 4.2.
2. Section 2.5. What could you do to improve the response of the system. How did you achieve that response. Comment on how you modified your controller to arrive at those results.

IV. HOME ANALYSIS
1. Bode plots and comparison for section 5.1
2. Plots and answer for section 5.2

V. CONCLUSIONS
1. Summarize the results obtained in this experiment.

Guide V2.0 Prepared by Daniel Posada
Embry-Riddle Aeronautical University
AE443/AE495 Laboratory

Quanser Inc.
119 Spy Court
Markham, Ontario
L3R 5H6
Canada
info@quanser.com
Phone: 1-905-940-3575
Fax: 1-905-940-3576

Printed in Markham, Ontario.

For more information on the solutions Quanser Inc. offers, please visit the web site at:
http://www.quanser.com